CSO Lab 5: Getting to Know the 8051 MCU

Question: Write a program that will perform a binary count from 00h to FFh.

```
ORG 0000h

START:
    MOV A, #00h

MAIN_LOOP:
    ; Check if Button P0.0 is pressed (Switch A on Keypad)
    JB P0.0, MAIN_LOOP

    ; Output and Increment
    MOV P1, A
    INC A           ; FFh + 1 will naturally roll over to 00h

    ; Repeat
    SJMP MAIN_LOOP

END
```

Code Explaination:

1. The Initialization

- **ORG 0000h**: This tells the microcontroller that the program starts at memory address 0000h. This is the standard "Reset Vector" where the 8051 looks for instructions when it first powers on.
- **START: MOV A, #00h**: This label marks the beginning of our setup. We move the value 00h (zero) into the **Accumulator (A)** register. This register acts as our "counter variable."

2. The Control Logic (The Button)

- **MAIN_LOOP:**: This is a label we use as a "return point" so the program can repeat indefinitely.
- **JB P0.0, MAIN_LOOP**: This is a **Jump if Bit is Set** instruction.
    - It looks at **Pin 0 of Port 0 (P0.0)**.
    - In the 8051, pins are "High" (1) by default. When you click the switch in your keypad, it pulls the pin "Low" (0).
    - **Logic:** If the button is **NOT** pressed (pin is 1), it jumps back to MAIN_LOOP. It effectively "traps" the program here until you press the button.

3. The Output and Action

- **MOV P1, A**: Once the button is pressed, the program moves to this line. It copies the current value of the Accumulator into **Port 1 (P1)**. This is what turns your LEDs on or off.
- **INC A**: This increments the Accumulator by 1.
    - Since the Accumulator is an 8-bit register, it counts from 00h (0) to FFh (255).
    - When it hits FFh and increments again, it naturally **rolls over** back to 00h automatically.

4. The Loop and Termination

- **SJMP MAIN_LOOP**: This is a **Short Jump**. It sends the program back to the MAIN_LOOP label to check the button again.
- **END**: This is a directive for the assembler to stop reading the file. It does not turn into a machine instruction.

Demo Video
https://drive.google.com/drive/folders/1rnfQfxfr03Xsfzd2DmIeLebdBVp3SIGy